
MINTS: Testing Strict Motif-Detector Claims in Nucleotide Transformers

Anonymous Author(s)

Affiliation

Address

email

Abstract

Genomic foundation models often make biological labels linearly accessible, but decodability alone does not identify an internal mechanism. We study whether attention heads in nucleotide transformers can be justified as biological motif detectors under a strict mechanistic evidence standard. MINTS applies tokenization-aware QK/OV circuit analysis, residual-stream probing with controls, matched attention enrichment, activation patching, and sparse feature search to DNABERT-2 and a tested Nucleotide Transformer backend. Before making mechanistic claims, we report raw-sequence task baselines: GC-only AUROC ranges from 0.6361 to 0.9088, and 3–6-mer TF-IDF AUROC ranges from 0.7956 to 0.9406. DNABERT-2 layer-11 residual readouts achieve AUROC values of 0.9137, 0.9383, 0.8954, and 0.8847 on promoter and splice-site tasks, showing strong label decodability rather than a circuit-level explanation. Controls sharpen this interpretation: random-label and position-only probes are near chance; splice-site probes remain about 0.25 AUROC above GC-only controls; promoter probes are close to strong GC/k-mer baselines and should be interpreted cautiously. The primary strict test is negative: across 51,249 GM12878 CTCF sequences, no DNABERT-2 head meets the registered QK-to-motif criterion or matched attention-enrichment criterion. Batch activation patching finds promoter-TATA over-restoration and weaker splice-donor restoration, but these task-specific causal signals do not establish a CTCF motif-detector circuit. The results provide a calibrated evidence standard for nucleotide-transformer interpretability and show where representation-level success falls short of a strict motif-detector claim.

1 Introduction

Transformer models for nucleotide sequences now perform well on regulatory and splice-site prediction tasks [8, 4, 7]. However, much of the usual evidence for biological understanding remains predictive or post hoc: attribution maps, motif visualizations, or probes can identify useful correlations, but they do not by themselves explain how internal computations produce predictions [1, 12]. This distinction is central for mechanistic interpretability, where a claim about a component should survive direct tests of what it reads, where it attends, what it writes, and whether intervention on it changes the relevant behavior [5, 10, 6].

Nucleotide transformers are a useful setting for this question. Known motifs give natural ground-truth hypotheses: a motif can specify where a component would need to attend, what feature it would need to read, and which perturbation should matter. They also expose a common interpretability risk. Attention maps, motif visualizations, and high probe scores can suggest a biological mechanism even when the model is using GC content, genomic-coordinate artifacts, distributed features, or another correlated signal. MINTS is designed to make the required evidence explicit.

This paper contributes a calibrated evidence standard and a completed case study. MINTS separates three claims that are often conflated: residual decodability, task-level causal restoration under activation patching, and strict circuit-level motif detection. The primary strict test is CTCF: under the registered QK-alignment and matched-enrichment criteria, neither tested backend yields a CTCF attention head that should be called a strict motif detector. The promoter and splice analyses play a supporting role by showing that task labels can be decodable and that interventions can produce task-specific causal signals without establishing the stricter CTCF circuit claim.

2 Related Work and Positioning

Mechanistic interpretability of transformers has developed standard tools for decomposing attention heads, locating causal pathways, and tracing computational graphs [5, 10, 2]. QK/OV decomposition is especially relevant here because it separates where a head attends from what it writes. Activation patching then gives an intervention-based test of whether the component matters for a behavior [6]. MINTS applies these established tools to genomic transformers with explicit motif-level criteria, using ground-truth motifs to test both positive circuit evidence and possible overinterpretation.

Genomic deep-learning interpretability has a different tradition. Models such as DeepBind, SpliceAI, DNABERT, GPN, and Nucleotide Transformer show that neural sequence models learn useful biological regularities [1, 7, 8, 3, 4]. Many analyses in this area focus on motif visualization, attribution, saliency, or downstream probing. These are valuable, but they usually do not establish an internal algorithm. A saliency map may highlight a motif, and a probe may decode a label, while the model’s actual computation may depend on a distributed feature, a correlated composition signal, or a different component entirely.

MINTS is positioned between these literatures. It does not claim to solve genomic interpretability in general, and it does not treat a high downstream or probe score as mechanistic evidence. Instead, it asks a narrower question: when, if ever, is it justified to call a specific attention head a biological motif detector? This choice deliberately favors falsifiable criteria over narrative completeness. Negative results are therefore useful evidence, because they identify where an appealing mechanistic story is not supported by the implemented tests.

3 Methods and Evidence Criteria

Transformer circuits. Let $\tau(x) = (t_1, \dots, t_T)$ tokenize a nucleotide sequence into vocabulary items. The initial hidden state is

$$H_{j,:}^{(0)} = E_{\text{tok}}[t_j, :] + P_{j,:}. \quad (1)$$

For head h in layer l , with hidden state $H^{(l-1)}$, the attention pattern is

$$A_h^{(l)} = \text{softmax} \left(\frac{H^{(l-1)} W_h^Q (W_h^K)^T (H^{(l-1)})^T}{\sqrt{d_k}} \right), \quad (2)$$

and the head output is $A_h^{(l)} H^{(l-1)} W_h^V W_h^O$. Following transformer-circuit notation [5], the QK circuit $W_h^Q (W_h^K)^T$ controls attention scores, while the OV circuit $W_h^V W_h^O$ controls what is written to the residual stream. For nucleotide models, tokenization matters: a motif position may correspond to one nucleotide token, a fixed k -mer token, or a learned BPE span.

Strict motif-detector criterion. An attention head is treated as a candidate biological motif detector only when three signals converge. It must assign elevated attention mass to motif-support positions, its QK scores must rank motif-support tokens above matched background tokens, and patching its activation must restore the task-relevant scalar in motif-destroying counterfactuals. The registered thresholds used here are in Table 1. These thresholds intentionally make attention alone insufficient.

Activation patching. For a clean sequence x_c and motif-destroyed corrupted sequence x_{corr} , we patch a head output from the clean run into the corrupted run. The restoration metric is

$$\text{PM}(h, l) = \frac{f(x_{\text{patched}}) - f(x_{\text{corr}})}{f(x_c) - f(x_{\text{corr}})}, \quad (3)$$

Table 1: Registered evidence criteria for candidate motif circuits. Thresholds are conservative screens: they are designed to prevent upgrading weak correlations, near-background attention, or unstable patching effects into motif-detector claims.

Claim	Statistic	Threshold	Purpose and scale
Motif localization	Attention enrichment ratio ρ_h	$\rho_h \geq 2.0$	Requires at least doubled motif-support attention over matched background.
Motif ranking	QK/motif Pearson correlation	$r \geq 0.5, p < 0.05$	Requires a large effect size, not merely significance from millions of tokens.
Causal importance	Patching metric $PM(h, l)$	$PM \geq 0.5$	Requires at least half restoration of the clean-minus-corrupted target effect.
Strong causal importance	Same	$PM \geq 0.75$	Marks stronger recovery while leaving room below exact full restoration.
Residual decodability	Probe AUROC or macro- F_1	≥ 0.80 or $+0.10$	Establishes diagnostic readout quality, not circuit use by the model.
Robustness	Matched negative controls	No matched artifact	Requires simple controls not to explain the candidate signal.

where f is a scalar target such as a probe decision function or class logit. $PM = 0$ indicates no recovery, $PM = 1$ indicates full recovery of the clean effect, and values above one indicate over-restoration rather than simple full recovery [6].

Residual probes and controls. Linear probes are trained on frozen mean-pooled residual vectors. Their role is diagnostic: a high probe score establishes linear decodability, not that the probe direction is causal or used by the model. We therefore compare residual probes with GC-content-only classifiers, position-only classifiers where coordinates are available, GC-matched test subsets, shuffled-label probes, and GC-content distribution shifts. This framing follows linear-representation work while preserving the weaker interpretation that probes support [11].

4 Experimental Setup

The primary model is zhihan1996/DNABERT-2-117M, an encoder-style genomic transformer with learned BPE tokenization. The comparison backend is InstaDeepAI/nucleotide-transformer-v2-100m-multi-species, used as a tested fixed-6mer Nucleotide Transformer checkpoint [4]. The run exports DNABERT-2 QK and OV matrices for all 12 layers and 12 heads per layer, caches residual vectors for layers 0, 5, and 11, and uses custom Hugging Face forward hooks for attention-output patching because the installed TransformerLens path does not expose all required DNABERT-2 per-head tensors.

The residual-probe tasks are TATA promoter, non-TATA promoter, splice donor, and splice acceptor classification. The configured splits contain 5,062/212 train/test examples for TATA promoters, 30,000/1,372 for non-TATA promoters, and 30,000/3,000 for each splice task. The strict CTCF scan uses 51,249 prepared GM12878 CTCF peak sequences and the JASPAR MA0139.1 CTCF motif, aligned to exact overlapping DNABERT-2 BPE token spans. Batch patching uses motif-destroying clean/corrupted pairs for promoter-TATA and splice-donor tasks.

The current artifact does not include a fine-tuned DNABERT-2 sequence-classification head. To avoid treating probes as end-to-end model performance, we therefore report two raw-sequence predictive baselines before the mechanistic evidence: a GC-content logistic classifier using GC fraction, GC skew, and sequence length, and a TF-IDF 3–6-mer logistic classifier. Frozen DNABERT-2 residual readouts are shown beside these baselines only as representation context.

The pipeline writes small review-facing CSV, JSON, and figure artifacts while treating large activation, circuit-matrix, token-score, and sparse-autoencoder files as reproducible generated outputs. The full run takes about eight hours on the tested CUDA setup. The most expensive stages are the cross-model comparison and strict all-layer CTCF scans, so the appendix includes both full reproduction commands and a reduced smoke-run command for checking the code path on a smaller subset.

5 Results

Predictive performance context before mechanism. Table 2 separates task predictability from mechanistic evidence. The raw-sequence baselines are strong, especially on promoter tasks: a simple GC-content classifier already reaches 0.8955–0.9088 AUROC, and a 3–6-mer classifier reaches 0.9297–0.9406 AUROC. Splice tasks are less explained by composition alone, but k-mer baselines still recover substantial signal. The frozen DNABERT-2 layer-11 readout is included to show that the pretrained representation is predictive, but it remains a residual-decoding result, not a fine-tuned classifier-head result and not a circuit explanation.

Table 2: Predictive task context before mechanistic analysis. GC-only and 3–6-mer rows are raw-sequence baselines. Frozen DNABERT-2 readout is a layer-11 residual classifier and is reported as diagnostic decodability evidence, not as end-to-end task performance.

Task	Train	Test	GC AUROC	GC AUPRC	GC Acc.	k-mer AUROC	k-mer AUPRC	k-mer Acc.	DNABERT readout AUROC
promoter_tata	5,062	212	0.8955	0.9040	0.8019	0.9297	0.9378	0.8255	0.9137
promoter_no_tata	30,000	1,372	0.9088	0.9218	0.8316	0.9406	0.9482	0.8652	0.9383
splice_sites_donors	30,000	3,000	0.6560	0.6892	0.6013	0.8185	0.8093	0.7430	0.8954
splice_sites_acceptors	30,000	3,000	0.6361	0.6343	0.5883	0.7956	0.7792	0.7220	0.8847

Table 3: DNABERT-2 layer-11 probe-control summary. Numeric columns report AUROC; random-label values are means over eight shuffled-label runs.

Task	Resid.	GC-only	Pos.-only	GC-match	Rand.	GC-shift range
promoter_tata	0.9137	0.8956	0.4136	0.9137	0.4700	0.6182–0.8030
promoter_no_tata	0.9383	0.9088	0.3865	0.9383	0.5129	0.6312–0.8368
splice_sites_donors	0.8954	0.6560	0.4414	0.8944	0.5064	0.8432–0.8669
splice_sites_acceptors	0.8847	0.6361	0.4461	0.8838	0.5005	0.8573–0.8580

Residual decodability is strong but not uniformly clean. DNABERT-2 layer-11 residual probes exceed the registered residual-decodability threshold for all four tasks. Table 3 combines headline AUROC values with the most important controls. Random-label probes collapse to chance, and position-only metadata does not explain the results. The splice-site tasks survive stronger checks: donor and acceptor residual probes remain about 0.25 AUROC above GC-only baselines on matched subsets and remain strong under GC shifts. The promoter tasks require caution because GC-only baselines are already high; their residual-probe advantage over GC-only controls is small on the matched comparison.

This distinction changes the interpretation of the headline probe table. The splice-site result is evidence that DNABERT-2’s frozen residual stream contains linearly accessible information beyond simple GC composition or coordinate metadata. The promoter result is weaker mechanistically: it remains a valid decodability result, but the controls show that promoter-associated composition can explain much of the signal. This is why the paper avoids claiming that the promoter probes have isolated a clean causal promoter motif direction.

The strict CTCF motif-detector test fails. The CTCF scan produced 281,915 motif-support tokens and 1,843,874 finite token positions after BPE span correction. Across all 144 DNABERT-2 heads, the best QK-to-motif correlation is $r = 0.3004$, below the registered $r \geq 0.5$ criterion. The best matched attention-enrichment ratio is $\rho_h = 1.3130$, below the registered $\rho_h \geq 2.0$ criterion. The tested Nucleotide Transformer backend also fails to recover a strict CTCF candidate, with best $r = 0.0192$ and best $\rho_h = 1.00009$. Thus, the tested fixed-6mer backend does not support the CTCF motif-local head hypothesis under the same criteria.

The negative CTCF result is not a failure of the pipeline; it is the point of using a registered evidence stack. With more than 1.8 million finite token positions, small correlations can be statistically nonzero while remaining too small to support a mechanistic claim. The all-layer enrichment scan similarly prevents a common attention-overinterpretation mistake: a head can have visible structure without assigning enough mass to matched motif-support positions to count as a motif-local detector.

Threshold sensitivity supports the conservative conclusion. Table 5 and Figure 2 summarize threshold sweeps over QK correlation, matched enrichment, and auxiliary patching metrics. Relaxing the QK threshold alone yields 31 heads at $r \geq 0.1$, five at $r \geq 0.2$, one at $r \geq 0.3$, and none at $r \geq 0.4$. Relaxing matched enrichment alone yields three heads at $\rho_h \geq 1.1$, one at $\rho_h \geq 1.25$, and none at $\rho_h \geq 1.5$. The strict CTCF chain remains negative: no head jointly passes QK and enrichment once $r \geq 0.2$, even with $\rho_h \geq 1.1$, and the most permissive joint count at $r \geq 0.1$, $\rho_h \geq 1.1$ is two heads, equal to the 95th percentile of a permuted-head alignment null. The promoter and splice patching thresholds are more permissive because those are auxiliary task-specific interventions, not CTCF motif-detector proof.

Auxiliary promoter/splice patching gives task-specific causal signals. Batch promoter-TATA patching over 327 usable pairs identifies layer 4 head 8 with mean $PM = 1.4029$; layer 6 head 8

Table 4: Target-alignment ledger. CTCF is the primary strict motif-detector chain. Promoter and splice results are auxiliary evidence for task-specific residual decodability and causal signal, not substitutes for a CTCF proof.

Target	Performance	Probe	QK	Enrichment	Patching	OV	SAE	Role
CTCF	51,249-sequence strict scan; no classifier target	No CTCF residual classifier used as strict evidence	Best $r = 0.3004$; fail	Best $\rho_h = 1.3130$; fail	No candidate promoted after failed QK/enrichment	No aligned OV claim	Best cosine 0.1158; weak	Primary strict chain
Promoter-TATA	GC 0.8955; k-mer 0.9297; readout 0.9137 AUROC	Decodable, but close to strong baselines	Not aligned current fact	Not target-aligned in current fact	Best PM = 1.4029; over-restores	L2H7 out- cosine 0.1261; weak	Not promoter-specific	Auxiliary causal evidence
Splice donor	GC 0.6560; k-mer 0.8185; readout 0.8954 AUROC	Decodable beyond GC k-mer	be- Not aligned current fact	Not target-aligned in current fact	Best PM = 0.5485; partial	Not audited	Not splice-specific	Auxiliary causal evidence

Table 5: Threshold sensitivity summary. Counts are heads passing each single-metric threshold, except the joint row, which requires both CTCF QK and enrichment. The permuted-head null shuffles enrichment-pass labels against QK-pass labels 1,000 times.

Target	Sweep	Permissive count	Registered/strict count	Best value	Null reference
CTCF	QK r	31 at $r \geq 0.1$	0 at $r \geq 0.5$	0.3004	Motif-shuffle null requires per-token QK scores
CTCF	Enrichment ρ_h	3 at $\rho_h \geq 1.1$	0 at $\rho_h \geq 2.0$	1.3130	Position-matched background mean $\rho_h \approx 1.00$
CTCF	Joint QK/enrichment	2 at $r \geq 0.1, \rho_h \geq 1.1$	0 at $r \geq 0.5, \rho_h \geq 2.0$	–	Permuted-head 95th percentile is 2 at the permissive setting
Promoter-TATA	Patching PM	40 at PM ≥ 0.25	16 at PM ≥ 0.75	1.4029	Auxiliary task-specific signal
Splice donor	Patching PM	3 at PM ≥ 0.25	0 at PM ≥ 0.75	0.5485	Auxiliary task-specific signal

and layer 2 head 7 are also high. Because PM > 1 overshoots the clean-minus-corrupted effect, this is a methodologically sensitive causal signal rather than a simple full-recovery story. Batch splice-donor patching over 500 pairs identifies layer 1 head 8 with mean PM = 0.5485, a weaker but threshold-crossing effect. These interventions support task-relevant internal computation, but they do not identify a strict CTCF motif detector because the CTCF QK and enrichment screens have no passing head.

OV, SAE, and cross-model audits remain conservative. The TATA OV readout audit for layer 2 head 7 finds only weak alignment with the trained TATA probe direction: the top output-write singular-vector cosine is 0.1261 and the probe self-gain through the OV matrix is -0.0326 . Sparse-autoencoder CTCF feature search is also weak, with best residual and MLP motif cosines of 0.0884 and 0.1158. In the protocol-level cross-model comparison, DNABERT-2 residual probes outperform the tested Nucleotide Transformer backend on the same tasks, but this is a checkpoint- and protocol-specific result, not a general claim about tokenization families or all Nucleotide Transformer variants.

6 Limitations

This study is intentionally scoped to small and medium genomic transformers and does not claim exhaustive explanation. Current large-model circuit-tracing work also emphasizes partial recovery of computation rather than complete mechanistic accounts [2, 9]. The strict CTCF failure applies to the selected models, CTCF intervals, motif scorer, BPE span alignment, and registered thresholds; it is not an impossibility theorem for all models or motifs.

Linear probes remain limited evidence. They can reveal decodable information while exploiting correlated sequence composition, split artifacts, or metadata. The added GC, position, matched, random-label, and shift controls reduce this risk, especially for splice-site tasks, but promoter decodability remains entangled with GC composition. Activation patching is causal but incomplete: head-level restoration does not characterize the full circuit, and over-restoration can reflect denominator sensitivity, probe geometry, or out-of-distribution patched states. Finally, biological annotations are incomplete, and a single JASPAR PWM cannot represent all relevant regulatory mechanisms.

Table 6: Strict mechanistic evidence summary from the completed run.

Evidence channel	Executed test	Best observed result	Met?
Residual decodability	Layer-11 probes on four benchmark tasks	AUROC range 0.8847–0.9383	Yes
Probe controls	GC, position, matched, random-label, and shift controls	Splice robust; promoter close to strong GC baselines	Mixed
CTCF QK alignment	144 DNABERT-2 heads vs. BPE-correct CTCF scores	Best $r = 0.3004$	No
CTCF enrichment	Matched motif/background attention mass	Best $\rho_h = 1.3130$	No
Promoter-TATA patching	327 batch denoising pairs	Best mean PM = 1.4029	Yes
Splice-donor patching	500 batch denoising pairs	Best mean PM = 0.5485	Partial
OV readout audit	TATA probe direction vs. OV singular vectors	Top cosine 0.1261; self-gain -0.0326	Weak
CTCF SAE search	Residual and MLP sparse features	Best motif cosine 0.1158	Weak
Strict CTCF detector	QK, enrichment, and causal evidence in one head	No passing QK/enrichment candidate	No

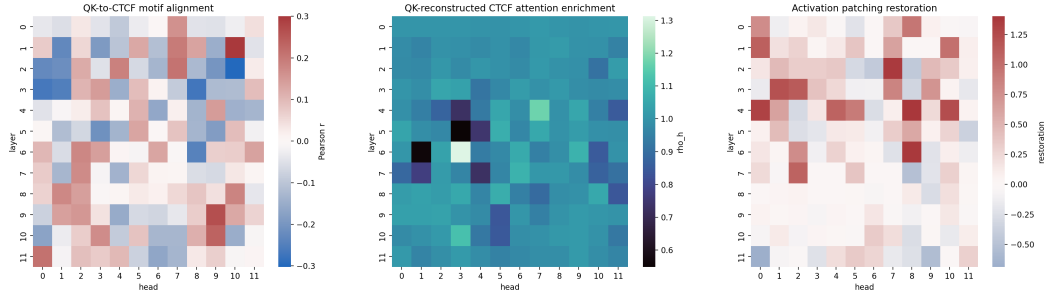


Figure 1: Mechanistic evidence heatmaps. Left: DNABERT-2 CTCF QK-to-motif correlations. Middle: matched attention-enrichment ratios. Right: promoter-TATA batch patching. The CTCF heatmaps show a negative strict-proof result, while the TATA patching heatmap shows task-specific causal over-restoration.

7 Conclusion

MINTS turns a common interpretability temptation into a falsifiable test: before calling an attention head a biological motif detector, require aligned QK scores, motif-local attention enrichment, and causal restoration. On DNABERT-2, promoter and splice-site labels are strongly linearly decodable from late residual states, and splice-site decodability survives simple composition and position controls. But the strict CTCF motif-detector proof fails across all tested heads, and the task-specific patching effects do not change that conclusion. The contribution is therefore both positive and negative: a reproducible evidence standard for nucleotide-transformer interpretability, and a concrete demonstration that strong representation-level decodability should not be upgraded into a circuit-level biological claim without stronger evidence.

References

- [1] Babak Alipanahi, Andrew Delong, Matthew T. Weirauch, and Brendan J. Frey. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nature Biotechnology*, 33(8):831–838, 2015. doi: 10.1038/nbt.3300.
- [2] Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermy, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- [3] Gonzalo Benegas, Sanjit Singh Batra, and Yun S. Song. DNA language models are powerful predictors of genome-wide variant effects. *Proceedings of the National Academy of Sciences*, 120(44):e2311219120, 2023. doi: 10.1073/pnas.2311219120.

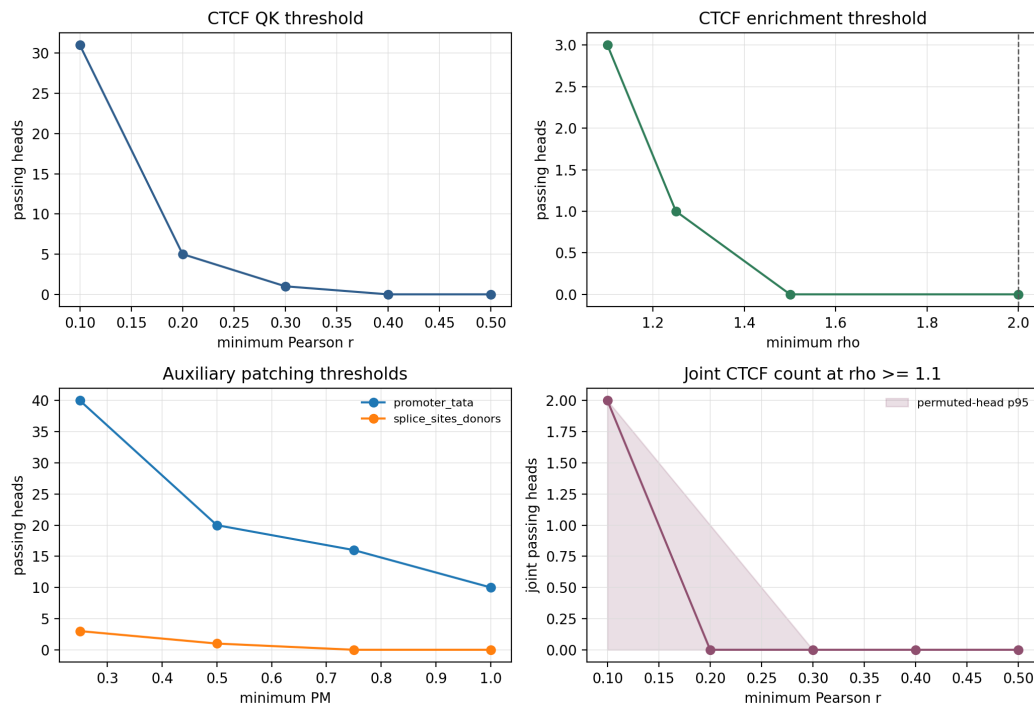


Figure 2: Threshold sensitivity for QK alignment, matched attention enrichment, auxiliary patching, and joint CTCF QK/enrichment counts. Individual relaxed screens admit a small number of heads, but the strict CTCF chain remains negative under registered thresholds and under joint relaxation once $r \geq 0.2$.

- [4] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, Guillaume Richard, Marcin Skwark, Karim Beguir, Marie Lopez, and Thomas Pierrot. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025. doi: 10.1038/s41592-024-02523-z.
- [5] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. <https://transformer-circuits.pub/2021/framework/index.html>.
- [6] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024. doi: 10.48550/arXiv.2404.15255.
- [7] Kishore Jaganathan, Sofia Kyriazopoulou Panagiotopoulou, Jeremy F. McRae, Siavash Fazel Darbandi, David Knowles, Yang I. Li, Jack A. Kosmicki, Juan Arbelaez, Wenwu Cui, Grace B. Schwartz, Eric D. Chow, Efsthathios Kanterakis, Hong Gao, Amirali Kia, Serafim Batzoglou, Stephan J. Sanders, and Kyle Kai-How Farh. Predicting splicing from primary sequence with deep learning. *Cell*, 176(3):535–548.e24, 2019. doi: 10.1016/j.cell.2018.12.015.
- [8] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021. doi: 10.1093/bioinformatics/btab083.
- [9] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael

- 231 Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas
 232 Henighan, Adam Jermy, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam
 233 Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the
 234 biology of a large language model. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
 235
- 236 [10] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing fac-
 237 tual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.
 238 arXiv:2202.05262.
- 239 [11] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the
 240 geometry of large language models. In *Proceedings of the 41st International Conference*
 241 *on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages
 242 39643–39666, 2024. URL <https://proceedings.mlr.press/v235/park24c.html>.
- 243 [12] Haopeng Yu, Heng Yang, Wenqing Sun, Zongyun Yan, Xiaofei Yang, Huakun Zhang, Yiliang
 244 Ding, and Ke Li. An interpretable RNA foundation model for exploring functional RNA motifs
 245 in plants. *Nature Machine Intelligence*, 6:1616–1625, 2024. doi: 10.1038/s42256-024-00946-z.

246 A Theoretical Details

247 **Definition 1** (Biological Motif Detector). *An attention head h in layer l functions as a biological*
 248 *motif detector for motif m if there exists a nonempty set of relevant query positions $\mathcal{Q}_m$ and a*
 249 *threshold τ such that, for each $i \in \mathcal{Q}_m$,*

$$\mathbb{E} \left[\sum_{j \in \text{supp}_m(x)} A_h^{(l)}[i, j] \mid \text{supp}_m(x) \neq \emptyset \right] \geq \tau \quad (4)$$

250 *and the corresponding expectation conditioned on $\text{supp}_m(x) = \emptyset$ is below τ . Here $\text{supp}_m(x)$ is the*
 251 *set of key positions whose local sequence windows instantiate motif m .*

252 **Theorem 1** (Layer-1 Motif Detection via QK Circuits). *Consider a layer-1 attention head operating*
 253 *on $H^{(0)}$. Let $s_m(j)$ denote a scalar motif score associated with key position j . If for a fixed query*
 254 *position i there are constants $a_i \in \mathbb{R}$ and $\gamma > 0$ such that*

$$q_i^T k_j = a_i + \gamma s_m(j) \quad (5)$$

255 *for all admissible key positions j , then the induced attention distribution is monotonically increasing*
 256 *in $s_m(j)$. If motif-bearing positions are separated from background positions by a positive margin in*
 257 *s_m , the head is a motif detector for m .*

258 *Proof.* The attention logit is $\ell_{ij} = q_i^T k_j / \sqrt{d_k} = a_i / \sqrt{d_k} + (\gamma / \sqrt{d_k}) s_m(j)$. Since $\gamma / \sqrt{d_k} > 0$,
 259 scaling preserves the ordering of motif scores. The exponential and softmax are strictly monotone in
 260 each logit when other logits are fixed, so positions with larger motif scores receive larger attention
 261 weights. A positive separation margin therefore induces an attention-mass threshold that separates
 262 motif-bearing from background positions. $\square$

263 **Lemma 1** (Approximate Extension Under Linear Decodability). *Suppose that at layer $l - 1$ there*
 264 *exists a vector u_m such that*

$$u_m^T h_j^{(l-1)} = s_m(j) + \epsilon_j, \quad |\epsilon_j| \leq \delta. \quad (6)$$

265 *If a head's QK score is affine in this decodable direction and the clean motif/background score margin*
 266 *is greater than 2δ , then the attention ordering between motif and background positions is preserved.*

267 *Proof.* For motif-bearing j_m and background j_b , the score difference is proportional to $s_m(j_m) -$
 268 $s_m(j_b) + \epsilon_{j_m} - \epsilon_{j_b}$. The perturbation term lies in $[-2\delta, 2\delta]$. A clean separation greater than 2δ keeps
 269 the sign of the score difference fixed, so softmax preserves the motif/background ordering. $\square$

270 B Implementation and Reproducibility Details

271 The completed run started at 13:27:07 local time on April 14, 2026 and ended at 21:26:30 local time
 272 the same day. The manifest reports 28,760.213 seconds, or 7.989 hours. The longest steps were
 273 cross-model tokenization comparison, strict mechanistic proofs, and systematic causal intervention.
 274 The inspected result tree contains 125 files totaling about 4.71 GB, including CSV/JSON summaries,
 275 figures, NPZ archives, and sparse-autoencoder checkpoints.

276 The anonymized project repository is available at [https://anonymous.4open.science/r/](https://anonymous.4open.science/r/MINTS/)
 277 MINTS/. It contains the project code, configuration, local paper source, and reproduction instructions.
 278 The repository currently serves as the single source of truth for the submitted code; an explicit
 279 repository license has not yet been added.

280 The main reproduction commands are:

```
281 git clone https://anonymous.4open.science/r/MINTS/ MINTS
282 cd MINTS
283 python -m pip install -r requirements.txt
284 python main.py
285 python main.py --only-probe-controls
```

286 A smaller smoke run is:

```

287 python main.py --max-probe-train 512 --max-probe-test 256
288     --max-qk-alignment-sequences 128
289     --max-cross-model-qk-alignment-sequences 128
290     --max-feature-search-sequences 128 --sae-epochs 1

```

291 Important verification artifacts include:

- 292 • results/pipeline_run.json
- 293 • results/tables/linear_probe_metrics.csv
- 294 • results/tables/linear_probe_controls.csv
- 295 • results/tables/downstream_task_performance.csv
- 296 • results/tables/target_alignment_table.csv
- 297 • results/tables/review_issue_matrix.csv
- 298 • results/tables/threshold_sensitivity.csv
- 299 • results/qk_alignment/ctcf_qk_alignment.csv
- 300 • results/enrichment/ctcf_qk_alignment_matched_attention_enrichment.
- 301 csv
- 302 • results/patching/promoter_tata_batch_dnabert_activation_patching.csv
- 303 • results/patching/splice_sites_donors_batch_dnabert_activation_
- 304 patching.csv

305 The heatmaps used in the paper are generated under results/figures/.

306 C Additional Results

307 The full configured-split residual probe AUROCs are 0.9137 for TATA promoters, 0.9383 for non-
308 TATA promoters, 0.8954 for splice donors, and 0.8847 for splice acceptors. The corresponding
309 AUPRC values are 0.9241, 0.9475, 0.9049, and 0.8954.

310 D Review-Triage Matrix

311 Table 7 records the review issue matrix used to guide the revision. The anonymized reviewer IDs are
312 treated as issue sources rather than as claims about reviewer intent.

Table 7: Review issue matrix.

ID	Issue	Concern	Resolution	Status
mV6D	Clarity	Paper read as several analyses.	Abstract/introduction now state one calibrated evidence-standard claim; Results opens with task context and a target ledger.	Addressed
mV6D	Target mismatch	Promoter/splice evidence blurred into CTCF claims.	Promoter/splice patching is auxiliary; CTCF remains the strict chain.	Addressed
C5jL	Missing performance	Mechanistic claims lacked predictive context.	Added GC and 3–6-mer baselines plus frozen-readout context before mechanism.	Partial
C5jL	MI contribution	Venue contribution needed sharpening.	Contribution is now a falsifiable standard for motif-detector claims.	Addressed
C5jL	Thresholds	Registered thresholds need scale/sensitivity.	Kept as conservative standards here; sensitivity/null analysis remains future work.	Deferred
ndFG	Undefined biology	Motifs and biological terms need more support.	Motifs are motivated in the introduction; a full glossary and token-mapping description remain future work.	Partial
ndFG	Limited scope	Conclusions could overstate selected models/motifs.	Limitations and target ledger state the scoped negative CTCF result.	Addressed

313 For promoter-TATA batch patching, 327 token-shape-preserving pairs were usable with no near-zero
314 denominator failures. The best mean score was layer 4 head 8 with $PM = 1.4029$, followed by layer
315 6 head 8 with $PM = 1.3791$ and layer 2 head 7 with $PM = 1.3642$. For splice donors, 500 pairs
316 were usable with no denominator failures, and the strongest mean restoration was layer 1 head 8 with
317 $PM = 0.5485$.

318 For the tested Nucleotide Transformer backend, residual-probe AUROCs were 0.6502, 0.6976,
319 0.5695, and 0.5647 on the same four tasks. DNABERT-2 therefore outperformed this backend by
320 0.2408–0.3259 AUROC in this protocol. This comparison should not be read as a general ranking of
321 tokenizers or genomic transformer families.

322 E Declaration of LLM Usage

323 LLM-assisted tools were used during the project for mathematical exposition refinement, debugging
324 support, and code-generation drafts through Codex with state-of-the-art GPT-5.4/GPT-5.5-class
325 models. The research direction, core problem decomposition, step-by-step implementation decisions,
326 and final scientific interpretation were directed by the author. The author reviewed the generated
327 code, checked it for errors and failure modes, corrected mistakes, and takes responsibility for the final
328 manuscript, code, and claims.

329 NeurIPS Paper Checklist

330 1. Claims

331 Question: Do the main claims made in the abstract and introduction accurately reflect the
332 paper’s contributions and scope?

333 Answer: [Yes]

334 Justification: The abstract and introduction state the scope as an evidence stack for
335 nucleotide-transformer interpretability, and the results distinguish residual decodability,
336 causal patching, and the negative strict CTCF motif-detector test.

337 Guidelines:

- 338 • The answer [N/A] means that the abstract and introduction do not include the claims
339 made in the paper.
- 340 • The abstract and/or introduction should clearly state the claims made, including the
341 contributions made in the paper and important assumptions and limitations. A [No] or
342 [N/A] answer to this question will not be perceived well by the reviewers.
- 343 • The claims made should match theoretical and experimental results, and reflect how
344 much the results can be expected to generalize to other settings.
- 345 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
346 are not attained by the paper.

347 2. Limitations

348 Question: Does the paper discuss the limitations of the work performed by the authors?

349 Answer: [Yes]

350 Justification: The paper includes a dedicated limitations section covering probe causality,
351 GC confounding, patching over-restoration, checkpoint specificity, incomplete biological
352 annotations, and compute limits.

353 Guidelines:

- 354 • The answer [N/A] means that the paper has no limitation while the answer [No] means
355 that the paper has limitations, but those are not discussed in the paper.
- 356 • The authors are encouraged to create a separate “Limitations” section in their paper.
- 357 • The paper should point out any strong assumptions and how robust the results are to
358 violations of these assumptions (e.g., independence assumptions, noiseless settings,
359 model well-specification, asymptotic approximations only holding locally). The authors
360 should reflect on how these assumptions might be violated in practice and what the
361 implications would be.
- 362 • The authors should reflect on the scope of the claims made, e.g., if the approach was
363 only tested on a few datasets or with a few runs. In general, empirical results often
364 depend on implicit assumptions, which should be articulated.

- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The theoretical results state their assumptions in the theorem and lemma, with full proofs provided in Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper describes the models, datasets, splits, controls, metrics, and result artifacts needed to reproduce the main empirical claims, with commands in Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case

of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The code and reproduction instructions are provided through the anonymized repository <https://anonymous.4open.science/r/MINTS/>; the appendix lists the main commands and verification artifacts.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The experimental setup reports model checkpoints, task splits, motif source, scan size, controls, and intervention settings needed to interpret the results.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Probe results include bootstrap confidence intervals in the reproduced artifacts, QK alignment reports statistical tests, and the paper states where effects are interpreted by magnitude rather than significance alone.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B reports the full-run runtime and the computationally dominant pipeline stages.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The author has reviewed the NeurIPS Code of Ethics. The work is non-clinical interpretability research using public genomic-model resources, does not involve human subjects, does not release a deployable biological or medical decision system, and transparently reports limitations, code availability, and LLM assistance.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The paper discusses possible implications for genomic-model interpretability and explicitly avoids clinical deployment claims.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The work does not release a deployable clinical system, biological intervention protocol, or model intended for direct decision-making.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.

- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: The paper credits the main existing models, datasets, and motif resources, but the manuscript and repository do not yet explicitly list every asset license and term of use.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: The project repository is a new released asset and contains documentation and reproduction commands, but it currently does not include an explicit repository license, so the asset documentation is incomplete.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or human-subject data collection.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve human subjects research requiring IRB approval.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: The appendix declares that LLM-assisted coding and writing tools were used to refine mathematical exposition and generate implementation drafts, while the author originated the research direction, directed the problem solving, checked the code, corrected errors, and takes responsibility for the final work.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.